

ETHAN MOON

616-295-5694 | ethmoon@umich.edu | www.linkedin.com/in/ethan-moon0108 | github.com/emoon0108 | emoon0108.github.io

Education

University of Michigan College of Engineering

Bachelor of Science in Engineering, Computer Science

Ann Arbor, MI

Expected May 2028

- **Relevant Coursework:** Computers & Programming, Discrete Math, Proof-based Linear Algebra, Multivariable Calculus, Differential Equations, Statistics, Physics

Technical Skills

Languages: TypeScript, JavaScript, Python, C++, SQL

Frameworks & Tools: React, React Native/Expo, Node.js, Hono, PostgreSQL, Drizzle ORM, OpenAPI/Orval, Docker, Git, Arduino, ESP 32, NASA GMAT, Computer Vision, Digital Twin, OpenFOAM/Paraview, Microsoft Office

Languages Spoken: English & Korean (Native/Bilingual); Spanish (Limited Working)

Experience

Lead Engineer & Head of US Team, Ody

May 2026 – Present

- Promoted from Software Engineering Intern to Lead Engineer and Head of US Team in September 2026 after ranking #1 intern in a class of 6 despite being the youngest team member.
- Shipped 107 merged PRs in 10 weeks across 6 product apps and the backend, owning TypeScript features end-to-end from PostgreSQL/Drizzle schemas and Hono APIs through OpenAPI/Orval clients to React/Expo interfaces.
- Delivered 4 merchant-facing platform features—including AI-assisted multilingual digital menus using openrouter API, email marketing campaign using resend API, and multi-location admin analytics—as secure, production-ready workflows.

Co-Author Researcher, Grand Valley State University

September 2025 – Present

- Developed and calibrated non-Newtonian, multiphase CFD models in OpenFOAM/RheoTool across 32 bioink deposition conditions, achieving 96% agreement with experimental data. Under mentorship of Dr. Abishek Balsamy Kamaraj
- Engineered a Dockerized, full-stack Python/JavaScript platform with VTK-to-CSV pipelines and interactive visualization to automate analysis of 15,300+ records across 51 timesteps per simulation.
- Programmed an AI for real-time bioprinting nozzle-anomaly detection using digital-twin modeling; co-authored a peer-reviewed ASEE conference paper and co-authoring a bioprinting journal paper on the results pending publication.

Summer Researcher, Grand Valley State University

Jun 2025 – August 2026

- **Orbital Dynamics** (Jun – Oct 2025): Simulated orbital transfers, rendezvous, and formation-flying maneuvers in Octave and NASA GMAT, implementing perturbation and fuel-efficient maneuver algorithms for satellite swarms and LEO constellations. Under mentorship of Professor Phillip Hittetpole
- **Aerodynamic Hysteresis** (Jun – Aug 2025): Designed and 3D-printed airfoil prototypes, ran wind-tunnel tests with Arduino-based sensors, and compared experimental data against CFD simulations to mitigate aerodynamic hysteresis under Professor Sanjivan Manoharan and Graduate student Brendan Workman.

Simulation & Robotics Engineering Intern, Casual Simulation

Feb 2024 – Jan 2026

- Used XP exchange and Meta Quest VR to engineer a rowing-performance simulation app adopted by a local high school rowing team, and iterated on Arduino-based control systems for a robotic dog, achieving a 21% increase in stability

Projects

No Spoilers (*nospoilers.xyz*) – Full-Stack Web App & Chrome Extension

Jun 2026 – Present

- Built a spoiler-free movie discovery platform with 114 API handlers and 41 PostgreSQL models spanning 37 user pages.
- Shipped a privacy-first Chrome extension with on-device spoiler classification and one-click watchlist syncing, passing 28/28 automated tests.

Founder & Lead Developer, Helping Hands Soccer Training

Mar 2022 – Apr 2026

- Built a full-stack web/mobile platform with location- and goal-based matching algorithms and progress-tracking dashboards to systemize coach-player pairing, growing the nonprofit to donate \$2,400 in scholarships.

ViolinTwin – AI Practice Coach, React Native/Expo

2026

- Prototyping a mobile app with a scaffolded native Swift/Kotlin analysis module that records practice audio and estimates pitch, rhythm, and note timing against imported MusicXML/MIDI targets for real-time feedback.

Smart Plate – Gamified Nutrition App for Children with Autism

Apr 2023 – Jun 2025

- Designed a gamified nutrition app with a simulated smart-plate integration and BLE-ready architecture, using low-stimulation UX to gradually expand food exposure; piloted with a child with severe selective eating and observed increased willingness to try new foods.

Honors & Awards

DECA State Champion · (2nd Place Engineering) GYM International STEM Innovation Challenge · Finalist, International Astronomy & Astrophysics Competition · Published in International STEM Research Journal · Dartmouth Book Award